Please note that this syllabus is subject to change and will be updated for the current semester.

DIGHUM 162
DATA SCIENCE FOR SOCIAL JUSTICE
TERM, YEAR

Course format: Online, Synchronous
3 Semester Credits
Class #

Course Description

This interdisciplinary course equips students with essential tools and methods for data analysis while introducing critical frameworks to design projects with a focus on social impact. By combining technical skills with ethical considerations, the course challenges the "market-first" mentality prevalent in data science and fosters an understanding of how data-driven solutions can advance social justice. Through a combination of hands-on workshops and guided discussions, students will explore topics such as bias in datasets, the ethics of machine learning, data surveillance, and the social implications of data gathering and usage. By examining these topics, students will learn to interrogate the social and ethical dimensions of data science practices.
The course is divided into two main components: workshops and discussions. In the workshops, students will engage with data analysis tools in Python, analyzing datasets of different modalities–numeric, textual, and geospatial–that impact different identity groups. The discussions will center around the ideas of leading thinkers in areas such as data feminism, data activism, ethics, indigenous technologies, and critical race theory. Topics such as data gathering, surveillance, bias in machine learning, and the ethical dimensions of data science will be explored.

The course leverages datasets, including the COMPAS Recidivism dataset, the Opportunity Atlas for health outcomes, and textual data from Reddit containing linguistic biases. These datasets allow students to explore the intersections of social justice and data science. Throughout the course, students will apply their technical skills to analyze disparities in outcomes and in doing so, develop a critical understanding of how structural inequalities are reflected and perpetuated in data.

Prerequisites

No prior programming knowledge is expected, and students are not evaluated on their coding skills. However, the course does utilize Jupyter Notebooks with Python code, providing an opportunity to enhance coding skills while gaining a deeper understanding of computational techniques and approaches.

Course Learning Objectives

Knowledge and Understanding:

- Demonstrate knowledge of critical data theory and methods, particularly those related to social justice and bias in datasets.
- Understand the epistemological challenges and ethical pitfalls in data science, focusing on real-world datasets of different modalities, including numeric data for recidivism (COMPAS), geospatial data for health outcomes (Opportunity Atlas), and textual data reflecting linguistic biases (Reddit).
- Gain familiarity with frameworks such as data feminism, critical race theory, and data activism to evaluate how structural inequalities manifest across numeric, textual, and geospatial datasets.

Applying Knowledge and Understanding:

- Explore positionality as scholars engaging with data science in the context of criminal justice, health disparities, and cultural norms reflected in social media data.
- Apply statistical methods and data cleaning techniques to numeric, textual, and geospatial datasets, uncovering patterns of bias and inequality, such as disparities in sentencing, health outcomes, and linguistic norms.
- Develop technical skills using Python to analyze datasets of different modalities, addressing systemic disparities and proposing socially impactful solutions based on data-driven insights.

Making Judgments:

- Critically assess the biases, norms, and assumptions embedded in computational tools, such as machine learning models for recidivism prediction, geospatial analysis, and language models.
- Evaluate the implications of predictive algorithms in perpetuating racial, gender, and socio-economic inequalities using the COMPAS dataset, Opportunity Atlas, and Reddit data to explore the social justice implications of algorithmic decision-making across domains.
- Reflect on the ethical dimensions of data gathering and analysis, applying critical perspectives to identify injustices in criminal justice, public health, and social media data practices.

Communication:

- Present data analysis and research findings in group projects and individual digital essays, addressing critical data ethics and equity concerns.
- Engage in weekly written reflections that connect technical practices with secondary literature and case studies, such as racial disparities in recidivism, geospatial health inequities, and linguistic bias in social media platforms.
- Communicate effectively through visualizations and narratives linking data analysis to broader issues of social justice, equity, and ethical data practices.

Skills:
- Critically analyze numeric, textual, and geospatial datasets using Python, employing advanced data cleaning, visualization, and statistical modeling techniques.
- Construct visualizations that effectively illustrate disparities in criminal justice outcomes, health inequities, and linguistic biases, and connect these findings to broader social and ethical critiques.
- Develop proficiency in Python-based tools, including Jupyter Notebooks, while applying critical data perspectives to datasets of varying modalities.

Instructor Information and Communication

Course Instructor:
EMAIL:
Communication Preferences: *Instructor will update*

Graduate Student Instructors (GSIs)
TO BE USED ONLY IF GSIs NEEDED
While the instructor will interact with the whole class and will oversee all activities and grading, as well as being available to resolve any issues that may arise, the GSIs will be your main point of contact. Your GSIs are responsible for assisting you directly with questions about assignments and course requirements. The GSIs will also facilitate ongoing discussion and interaction with you on major topics in each module.
- Name of GSI | email

Discussion Sections
REMOVE IF NO GSI Your GSI will lead weekly discussion sections via Zoom. [Instructor to add other details about discussion sections and attendance policy.]

Office Hours
The course instructor and GSIs will offer office hours via Zoom by appointment. Links to the appropriate Zoom meeting will be available in bCourses.

Course Help
You’re not alone in this course; the instructor is here to support you as you learn the material. It's expected that some aspects of this course will take time to grasp, and the best way to grasp challenging material is to ask questions.

To ask a question, please use the discussion board on the course website. The instructor and GSIs will monitor this discussion forum, but you should also feel free to answer questions posted by other students. You can also reach out to the course staff in office hours, during live discussion sections, and/or via email.

Students with Disabilities

If you require course accommodations due to a physical, emotional, or learning disability, contact UC Berkeley's Disabled Students' Program (DSP). Notify the instructor and GSI through course email of the accommodations you would like to use. You must have a Letter of Accommodation on file with UC Berkeley to have accommodations made in the course.

UC Berkeley is committed to providing robust educational experiences for all learners. With this goal in mind, we have enabled the ALLY tool for this course. You can now automatically generate “Alternative Formats” for course files and bCourses Pages. Depending on the context, these formats can include Tagged PDF, HTML, BeeLine Reader, Electronic Braille, ePub, Immersive Reader, MP3, and translated versions. For more information, watch the video entitled, "Ally Tutorial for Students at UC Berkeley.”

Course Materials and Technical Requirements

Textbook / Required Materials
No textbook is required. All materials will be provided.

Additional Materials
You’ll find links to additional reading materials in bCourses.

Technical Requirements
This course is built on a Learning Management System (LMS) called Canvas (UC Berkeley’s instance of Canvas is called bCourses). You’ll need to meet these computer specifications to participate within this online platform.
bCourses: All course materials will be available on the “Data Science + Social Justice” bCourses webpage. You will receive access to this course before starting the program.

- **Slack.** Informal, day-to-day course communication will occur on our Slack Workspace. You can also use Slack to communicate with your peers, project group, and instructors. Slack provides a great way to quickly and publicly transmit information and get help. You will receive an invitation to join the Slack Workspace before the start of the workshop. We recommend downloading the Slack application to your computer if you do not already have it.

- **Jupyter Notebooks:** The Jupyter notebooks are available through Github. Optional, but recommended, is to create a GitHub account if you don't have one yet. The instructions to install Python and Jupyter, as well as the course materials, can be found in the GitHub repository under the “Installation Instructions” section. Please follow these instructions closely prior to the workshop.

- **Data:** Datasets will be made available on bCourses.

- **Live Meetings via Zoom:** The live meetings will be conducted over Zoom, Mon-Thu. During the Monday and Wednesday meetings, we will discuss the critical texts we've read for that week. You will have the opportunity to share research developments for your final projects. You will learn computational techniques and programming in Python on Tuesday and Thursday.

Technical Support
UC Berkeley provides technology support for graduate and undergraduate students that may prove valuable in this course, including free software (software.berkeley.edu), device lending (technology.berkeley.edu/step) and the student helpdesk (studenttech.berkeley.edu/techsupport).
If you're having technical difficulties please alert one of the GSIs immediately. However, understand that neither the GSIs, nor the instructor can assist you with technical problems. You must call or email tech support to resolve any technical issues.
To contact tech support, click on the "Help" button on the bottom left of the global navigation menu in bCourses. Be sure to document all interactions (save emails and transaction numbers).

Critical and Conceptual Pillars

1. **Anti-colonial + Science and Technology Studies (STS) Perspectives.** The dominant view in the data science and social justice literature often obfuscates non-technical and non-Western perspectives. Here, we strive to center these voices, filling significant gaps in current approaches.

2. **Algorithmic Bias**. Machine Learning models can result in discriminatory and harmful decisions reproduced at scale – especially impacting folks not well represented in the data gathering and model development process. This workshop includes seminal pieces from this new, emerging discipline.

3. **Problematizing Data**. Who gets counted, and who is rendered invisible? As a society, we use these datasets to justify policy, but we rarely question how the data has been manipulated. Here, we frame the data as a construct we must rigorously analyze.

4. **Problematize Data Science.** Positivism gives rise to quantitative methodology. This methodology involves the collection of 'scientific' data that is precise and based on measurement and is often analyzed using statistics with the intention that the findings be generalizable. Postpositivism is a metatheoretical stance that critiques and amends positivism and has impacted theories and practices across philosophy, social sciences, and various models of scientific inquiry. "Data Science" signifies a new positivist turn, and in this workshop, we will think critically about the term and its hegemonic impact in higher education and beyond.

Learning Activities and Assignments

This course emphasizes active engagement, participation, and completion over evaluative criteria, encouraging self-guided learning. The structure allows students to explore challenging new concepts while engaging critically with their peers on these ideas. While assessment focuses primarily on task completion, students are expected to engage seriously with the material and use the course as an opportunity to grow as scholar-activists.

Attendance and Participation-

Participation is 20% of the final grade and includes:

- Attending all class meetings and engaging in discussions (both in class and online via Slack or bCourses).
- Actively contributing to workshops, group work, and project critiques.
- Students who miss a class can make up participation points by submitting a 1-page reflection on the assigned reading within 48 hours of the missed session.

Weekly Readings

Students are expected to complete weekly readings before the Monday session. In most cases, a selection of readings will be provided. Students will be asked to engage in reflective exercises such as answering reflection questions or annotating the readings to prepare for live discussions.

Weekly Assignments

Students will complete smaller assignments that build toward their final project each week. These include group work during live sessions, collaborations outside of class, or individual tasks. They also include written reflections that connect technical practices with secondary literature and case studies, such as racial disparities in recidivism, geospatial health inequities, and linguistic bias in social media platforms. Assignments will be accessible through bCourses Modules.

Final Project

The course culminates in a two-part final project comprising group and individual components. Students will apply the techniques learned throughout the course to address research questions of their own design while working with peers and receiving support.

Final Group Project

The group component of the final project involves analyzing a dataset in groups of 4-5 students. During the live sessions of Week 6, each group will present their research path, starting with their research question and final product, followed by an explanation of their process. The focus will be on what makes their dataset and findings interesting.

Final Essay/Research Project (Individual Digital Essay)

For the individual component, students will submit a Jupyter Notebook digital essay synthesizing the research conducted in the group project. The essay is an individual effort, integrating students' work from the weekly assignments, thereby connecting the technical to the theoretical. It will be completed in three phases:

- Project Proposal: Due [INSERT DATE]
- Project Introduction: Due [INSERT DATE]
  By this point, students should have a draft of their introduction, providing an overview of the dataset and research questions.
- Project Draft: Due [INSERT DATE]
  The draft should be approximately 1,000 words, including a full introduction and a roadmap for the body and potential conclusions.
- Final Project: Due [INSERT DATE] by 11:59 PM Pacific Time.

All written work must follow MLA guidelines. See http://writing.wisc.edu/Handbook/Documentation.html for more information on these guidelines and examples of their correct use. All essays must contain pagination.

Grading

Your final course grade will be calculated as follows:

Category	Percentage of Grade	Notes
Attendance/Participation	20%	
Weekly Assignments/Reflections	30%	
Final Group Project	25%	
Final Essay/Research Project (Individual Digital Essay)	25%	

Strategies for Successful Learning

Take Care of Yourself

Do your best to maintain a healthy lifestyle this semester by eating well, exercising, getting enough sleep, and taking time to recharge your mental health. Taking time to care for yourself, and avoiding burnout, will help you achieve your academic, professional, and personal goals.

If you start to feel overwhelmed, be kind to yourself and reach out for support. Remember that seeking help is a courageous thing to do—for yourself and for those who care about you.

Support Resources include emotional, physical, safety, social, and other basic wellbeing resources for students. Academic resources can be found at the Student Learning Center and English Language Resource sites. Berkeley's Office of Emergency Management has resources to prepare for emergencies.

Course Policies

Attendance Policy

Students are expected to attend every class meeting, and should arrive on time, with cameras on, ready to participate. If students have a valid reason for missing class (illness, emergency, religious observance), they must inform the instructor before class to request an excused absence. More than two unexcused absences will result in a deduction of 5% per absence from the participation grade.

Collaboration and independence

Reviewing lectures and reading materials can be an enjoyable and enriching thing to do with fellow students. This is recommended, especially when it comes to the Colab notebooks. However, weekly assignments are to be completed independently and should be the result of one's own independent work.

Late Work/Retake Policy

Late work is accepted but with penalties to encourage timely submission. Assignments may be submitted **up to 5 days** after the original due date. A **10% deduction per day** will be applied to late assignments. After 5 days, the assignment will no longer be accepted unless prior arrangements have been made.

Extenuating Circumstances: If an emergency arises (e.g., illness, family emergency), students should reach out **before the due date** (when possible) to discuss an extension.

Retake Policy

Retakes are allowed for major assessments (e.g., exams, projects) but not for daily homework or participation-based work. Students must request a retake within one week of receiving their grade. Students may retake one major assessment per semester. The new grade will be an average of the original and retake scores to reflect both effort and improvement.

Documentation for Late Work

- For **1-2 days late**, no formal documentation is required.
- For **3+ days late**, students should provide a brief written explanation.
- For **extensions beyond 5 days**, official documentation (e.g., a doctor's note or university verification) may be required

Academic Integrity

You're a member of an academic community at one of the world's leading research universities. Berkeley creates knowledge that has a lasting impact in

the world of ideas and on the lives of others; such knowledge can come from an undergraduate paper as well as the lab of an internationally known professor. One of the most important values of an academic community is the balance between the free flow of ideas and the respect for the intellectual property of others. Scholars and students always use proper citations in papers; professors may not circulate or publish student papers without the writer's permission; and students may not circulate or post materials (handouts, exams, syllabi—any class materials) from their classes without the written permission of the instructor. Any test, paper or report submitted by you and that bears your name is presumed to be your own original work that has not previously been submitted for credit in another course unless you obtain prior written approval to do so from your instructor. In all of your assignments, including your homework or drafts of papers, you may use words or ideas written by other individuals in publications, websites, or other sources, but only with proper attribution. If you're unclear about the expectations for completing an assignment or taking a test or examination, be sure to seek clarification from your instructor or GSI beforehand. For additional information, read this UC Berkeley Library guide on How to Avoid Plagiarism.
As a member of the campus community, you're expected to demonstrate integrity in all of your academic endeavors and will be evaluated on your own merits. The consequences of cheating and academic dishonesty—including a formal discipline file, possible loss of future internship, scholarship, or employment opportunities, and denial of admission to graduate school—are simply not worth it. Read more about Berkeley's Honor Code.

Incomplete Course Grade
Students who have substantially completed the course but for serious extenuating circumstances, are unable to complete the remaining course activities, may request an Incomplete grade. This request must be submitted in writing to the GSI and instructor. You must provide verifiable documentation for the seriousness of the extenuating circumstances.
Refer to the Office of the Registrar's website for more information on the university's policy on Incomplete Grades.

End of Course Evaluation

UC Berkeley is committed to improving our online courses and instruction. Before your course ends, please take a few minutes to participate in the course evaluation. We are interested in your online learning experience, and your feedback will help us plan for the future and make improvements. The evaluation does not request any personal information, and your responses will remain strictly confidential. Information about the course evaluation will be made available in bCourses.

Course Outline

Readings and assignments will be updated on the course website.

Week 1: Introduction to Data Science for Social Justice

Monday

- **Reading:** Course Syllabus; Introduction to Data Feminism (D'Ignazio & Klein, 2020)
- **Assignments Due:** None

Tuesday

- **Reading:** Noble, S. (2018). "Algorithms of Oppression" (Chapter 1)
- **Assignments Due:** Reflection on Data Feminism (300 words)

Week 2: Problematizing Data and Algorithmic Bias

Monday

- **Reading:** Hoffmann, A. (2019). "Where Fairness Fails"
- **Assignments Due:** Reflection on algorithmic bias (300 words)

Tuesday

- **Reading:** Bolukbasi et al. (2016). "Man is to Computer Programmer as Woman is to Homemaker?"
- **Assignments Due:** Data exploration in Python (COMPAS dataset)

Week 3: Recidivism through Predictive Models

Monday

- **Reading:** O'Neil, C. (2016). "Weapons of Math Destruction" (Chapter 5)
- **Assignments Due:** Discussion post on recidivism and predictive models

Tuesday

- **Activity:** Building a basic logistic regression model (COMPAS dataset)
- **Assignments Due:** Python script for logistic regression

Week 4: Linguistic Bias and Social Media Data

Monday

- **Reading:** Caliskan et al. (2017). "Semantics Derived Automatically from Language Corpora"
- **Assignments Due:** Reflection on linguistic bias (300 words)

Tuesday

- **Activity:** Introduction to NLP and word embeddings (Reddit data)
- **Assignments Due:** Exploratory analysis of linguistic biases

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Week 5: Health Disparities through Geospatial Data

Monday

- **Reading:** Gil et al. (2019). "Urban Inequality and Health Outcomes"
- **Assignments Due:** Reflection on geospatial inequalities (300 words)

Tuesday

- **Activity:** Geospatial analysis using Opportunity Atlas
- **Assignments Due:** Data visualization of health disparities

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Week 6: Final Project Development

Monday

- **Reading:** Review of key course concepts
- **Assignments Due:** Peer review of project proposals

Tuesday

- **Activity:** Data processing and modeling for final projects
- **Assignments Due:** Preliminary data analysis

Assignment Deadlines Overview

Assignment	Due Date
Reflection on Data Feminism	Week 1, Tuesday
Reflection on Algorithmic Bias	Week 2, Monday

Data Exploration of COMPAS Dataset	Week 2, Tuesday
Discussion Post on Recidivism Models	Week 3, Monday
Logistic Regression Python Script	Week 3, Tuesday
Reflection on Linguistic Bias	Week 4, Monday
Word Embedding Analysis	Week 4, Tuesday
Reflection on Geospatial Inequalities	Week 5, Monday
Visualization of Health Disparities	Week 5, Tuesday
Peer Review of Project Proposals	Week 6, Monday
Preliminary Data Analysis	Week 6, Tuesday
Final Individual Digital Essay	Week 6, Tuesday

Module 1 and 2: Introduction
In the first week, students will set up Python and Jupyter Notebooks on their computers, ensuring they have the necessary tools to engage with the technical components of the course. Alongside technical onboarding, they will be introduced to foundational Python programming concepts, including variables, loops, and basic data structures. Conceptually, the course will open with a discussion of its critical and theoretical pillars, emphasizing anti-colonial perspectives, algorithmic bias, and the social implications of data science. These discussions will provide students with a strong ethical and theoretical framework to engage with data critically throughout the course.
Module 3: Criminal Justice Data and Bias
This module introduces students to the COMPAS Recidivism Dataset and criminal justice data from public records. Students will explore how these datasets highlight systemic biases in sentencing, recidivism rates, and other criminal justice outcomes. They will learn basic data cleaning techniques using Python's pandas library, focusing on handling missing data, renaming columns, and filtering datasets for analysis. Through this module, students will begin to understand how inequities are embedded in data systems and how computational tools can be leveraged to expose and critique these disparities.
Module 4: Recidivism through Predictive Models

Building on Module 3, students will analyze recidivism predictions using basic machine learning techniques like logistic regression and decision trees. They will investigate how predictive models built on biased data can perpetuate inequities and discuss the ethical implications of using such tools in criminal justice. Through hands-on exercises, students will implement these models in Python using scikit-learn and evaluate their performance, highlighting issues of fairness, accuracy, and societal impact.

Module 5: Linguistic Biases through Word Embeddings

This module explores biases in textual data by introducing students to word embeddings, focusing on how they capture and perpetuate cultural and societal norms. Using Python libraries like gensim or spaCy, students will analyze word vectors to identify associations reflecting gender, racial, and other biases in language. They will also learn to visualize these biases and critically assess the implications of using pre-trained language models in real-world applications. Discussions will connect these technical exercises to broader issues in digital humanities and computational linguistics, emphasizing the ethical dimensions of language technologies.

Module 6: Health Outcomes through Geospatial Analysis

In this module, students will analyze health outcomes using geospatial datasets like the Opportunity Atlas. They will learn to use Python tools such as geopandas and folium to visualize geographic data, exploring patterns of health disparities linked to socioeconomic and racial factors. Students will critically examine how structural inequities are spatially distributed and reflect on the implications of geospatial analyses for public policy and social justice. This module integrates data visualization techniques with discussions on systemic inequality, bridging computational methods with critical insights.

Module 7: Group Presentations and Individual Digital Essays

In the final week, students will present their group projects, showcasing the datasets they analyzed and the insights they gained about social justice and bias. Each presentation will include a discussion of the methodologies used, the challenges encountered, and the ethical questions raised by their findings. In addition to the presentations, students will submit individual digital essays synthesizing the research and analyses they conducted throughout the course. These essays, due as described above, and will demonstrate students' ability to integrate technical skills with critical frameworks from the weekly readings and reflections, thereby articulating the broader implications of their work.

Related Readings

Bolukbasi et al. (2016) – Man is to Computer Programmer as Woman is to Homemaker? Debiasing Word Embeddings
Caliskan et al. (2017) – Semantics Derived Automatically from Language Corpora Contain Human-Like Biases
Hoffmann (2019) – *Where Fairness Fails: Data, Algorithms, and the Limits of Antidiscrimination Discourse*
Dressel & Farid (2018) – The Accuracy, Fairness, and Limits of Predicting Recidivism
O'Sullivan et al. (2018) – Geographies of Inequality: Exploring Socioeconomic Disparities Using Geospatial Data
Gil et al. (2019) – Urban Inequality and the Spatial Distribution of Health Outcomes
Green (2021) – *Data Science as Political Action: Grounding Data Science in a Politics of Justice*